

MILLINGTON-MEMPHIS, TN, USA

WMO: 723284

Lat: 35.350N

Lon: 89.867W

Elev: 98

StdP: 100.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-6.0	-13.8	1.1	-5.0	-12.1	1.3	-3.6	9.5	9.0	8.6	8.4	1.9	10	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	37.5	27.1	36.2	26.4	34.1	25.5	28.6	35.6	27.7	33.9	26.8	32.8	2.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.0	32.9	26.1	21.8	31.6	25.1	20.4	30.4	92.3	35.2	88.5	33.9	84.5	32.1	31.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.1	5.8	-12.1	40.0	1.1	1.7	-12.9	41.2	-13.6	42.2	-14.2	43.2	-15.0	44.4		
			-12.7	29.6	0.9	1.3	-13.4	30.5	-13.9	31.3	-14.4	32.0	-15.1	33.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.8	4.8	6.5	12.9	17.3	21.6	26.5	27.6	27.2	23.0	16.8	11.5	6.0
	DBStd	9.28	6.16	5.77	5.64	4.64	4.07	2.62	2.54	3.05	3.19	4.60	4.87	5.13
	HDD10.0	543	184	128	36	4	0	0	0	0	0	3	41	147
	HDD18.3	1713	419	333	184	73	20	0	0	0	4	86	211	383
	CDD10.0	3044	24	29	126	224	359	494	545	532	389	214	85	22
	CDD18.3	1171	1	1	16	43	120	244	287	274	143	38	4	0
	CDH23.3	12561	0	3	119	401	1164	2754	3301	3114	1284	384	37	1
CDH26.7	5920	0	0	16	88	429	1385	1697	1661	522	117	4	0	
Wind	WSAvg	2.3	2.9	2.9	3.1	2.9	2.1	1.7	1.7	1.5	1.7	1.8	2.3	2.8
Precipitation	PrecAvg	1290	93	107	129	145	133	96	92	71	95	72	122	136
	PrecMax	1732	260	252	314	382	303	314	175	188	241	220	296	348
	PrecMin	849	21	36	17	42	22	10	25	3	5	0	36	22
	PrecStd	248	54	60	65	82	65	69	42	55	60	50	65	81
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.0	28.0	31.1	33.9	37.5	38.9	39.8	36.3	32.2	27.1	22.0
		MCWB	18.0	17.4	19.5	22.3	24.6	26.6	27.3	26.5	23.9	21.9	19.3	17.9
	2%	DB	19.8	21.2	27.0	28.8	32.4	36.1	37.3	37.7	33.0	29.7	23.9	18.9
		MCWB	16.6	16.5	18.6	20.0	23.6	25.9	27.8	27.0	23.3	21.0	17.4	16.1
	5%	DB	17.4	18.8	24.0	27.5	31.1	34.1	36.0	36.3	32.0	27.7	22.5	17.3
		MCWB	14.7	14.8	17.2	19.2	23.1	25.0	27.1	26.4	23.1	19.5	17.4	14.0
	10%	DB	15.1	16.8	22.4	26.1	29.0	32.9	33.8	34.0	29.7	25.4	20.9	14.1
		MCWB	12.5	12.5	16.1	18.7	22.0	24.7	26.0	25.9	22.0	18.3	16.1	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	19.2	20.7	22.9	26.3	28.1	29.6	29.7	26.4	24.5	21.7	18.7
		MCDB	21.2	22.1	26.4	28.8	32.4	34.9	36.7	38.0	33.6	29.2	24.7	21.1
	2%	WB	17.5	17.2	19.6	21.8	24.8	27.3	28.7	28.8	25.2	22.8	19.7	17.1
		MCDB	18.5	19.4	24.9	26.8	30.4	33.9	35.6	35.9	30.6	27.1	22.8	18.2
	5%	WB	15.2	15.4	18.5	20.7	23.9	26.4	27.8	27.6	24.3	21.5	18.1	14.6
		MCDB	17.2	18.5	23.2	25.3	29.2	33.0	34.2	33.5	28.9	25.2	21.1	16.9
	10%	WB	12.6	12.8	17.2	19.7	23.1	25.7	27.0	26.6	23.4	20.0	16.5	11.9
		MCDB	14.7	15.7	21.3	24.4	27.9	31.6	32.8	32.4	27.6	23.5	19.2	13.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.9	12.0	12.4	12.1	12.2	11.5	12.7	12.7	13.2	12.0	10.9
		MCDBR	12.1	12.8	13.2	13.6	13.7	13.5	13.8	14.9	15.0	15.9	14.9	12.6
		MCWBR	9.9	9.1	7.4	6.8	6.3	5.3	5.5	5.8	6.6	7.8	9.6	9.8
	5% WB	MCDBR	10.9	11.3	11.8	11.3	11.8	12.1	12.2	12.9	11.4	12.1	11.2	10.9
		MCWBR	9.7	8.9	7.3	7.2	6.1	5.4	5.5	5.8	5.9	7.4	8.9	9.8
Clear-Sky Solar Irradiance	taub	0.316	0.326	0.362	0.400	0.438	0.457	0.488	0.487	0.420	0.359	0.330	0.314	
	taud	2.495	2.463	2.388	2.278	2.233	2.188	2.139	2.122	2.304	2.455	2.488	2.515	
	Ebn at Noon	888	918	907	884	852	832	803	796	837	868	862	867	
	Edh at Noon	83	96	113	132	140	147	153	153	122	96	83	77	
All-Sky Solar Radiation	RadAvg	2.41	2.93	4.06	5.21	5.86	6.52	6.27	5.75	4.88	3.80	2.69	2.06	
	RadStd	0.20	0.38	0.35	0.35	0.37	0.40	0.31	0.35	0.35	0.45	0.28	0.21	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page